HCMAI 2026

Presenter:

- TA Tien Huy

Outline

- 1. Technique
- 2. Advice
- 3. QA + “Tâm sự”

1. Technique

1. Technique

❖ Problem Statement

- ***Known-Item Search (KIS)***:

  + A single video clip (a few seconds long) is randomly selected from the dataset and visually presented – this is known as **visual KIS (KIS-V).** The participants need to find exactly the single instance presented.

  + **Textual KIS (KIS-T)**, where instead of a visual presentation, the searched segment is described only by text given by the moderator (and presented as text via the projector).

  + Another variant of this is **KIS-C**, where the target scene is also textually described, but only with minimal details in the beginning. Further details are revealed after 60 seconds based on questions/chats from participants.

1. Technique

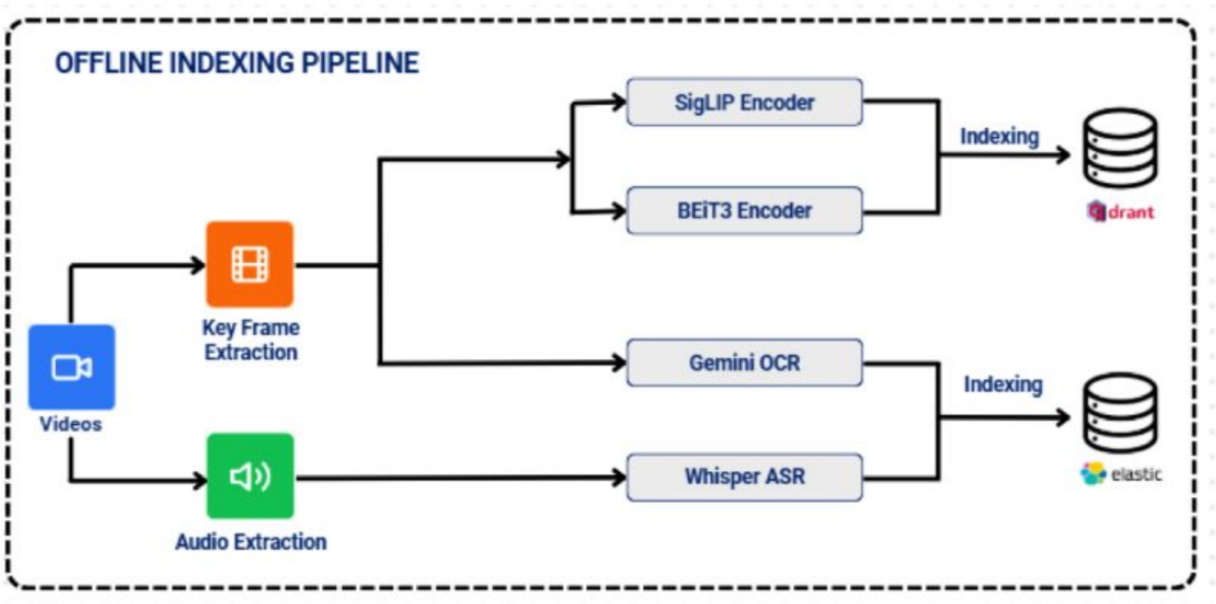
❖ Problem Statement

- **Visual Question Answering (VQA):** this task type asks specific questions about a particular video or video collection, which are intended to be submitted as a manually entered text (by a human). For example, ‘*How many nights do we see passing in the video until this segment?’*..... It requires manual inspection and interactive/ automatic exploration.

Cite: https://videobrowsershowdown.org/about-vbs/

1. Technique

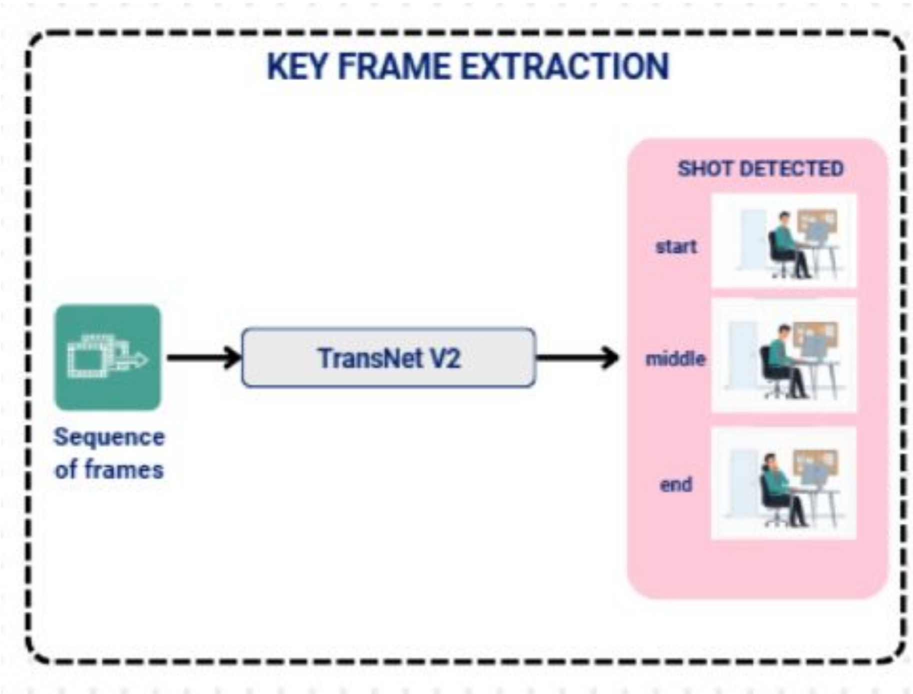
❖ Overall System



LINK

1. Technique

❖ Overall System



LINK

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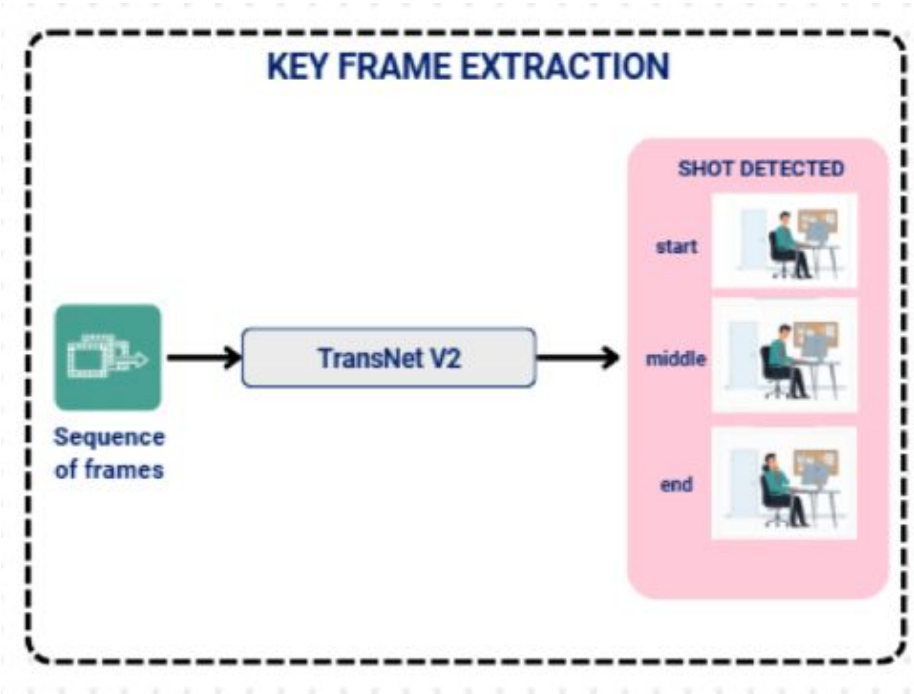
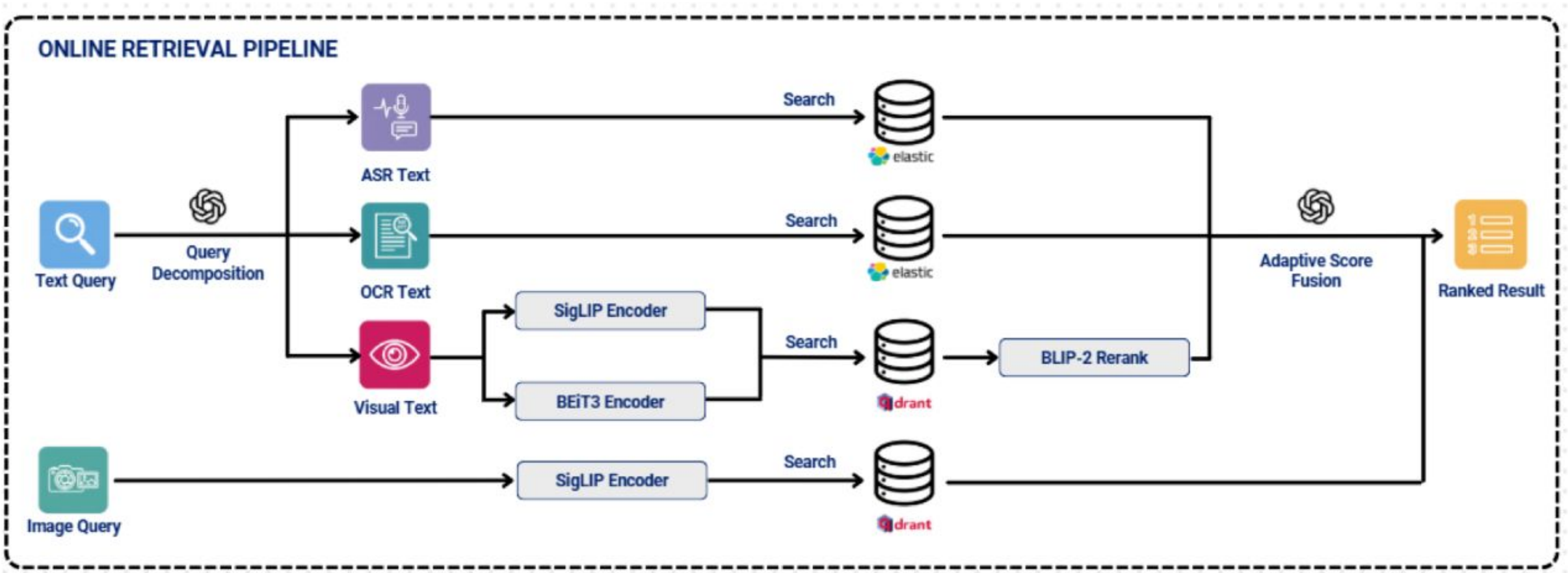


<u>With the increasing amount of data, filtering is becoming more and more important.</u>

[LINK]

1. Technique

❖ Overall System



LINK

AI VIETNAM
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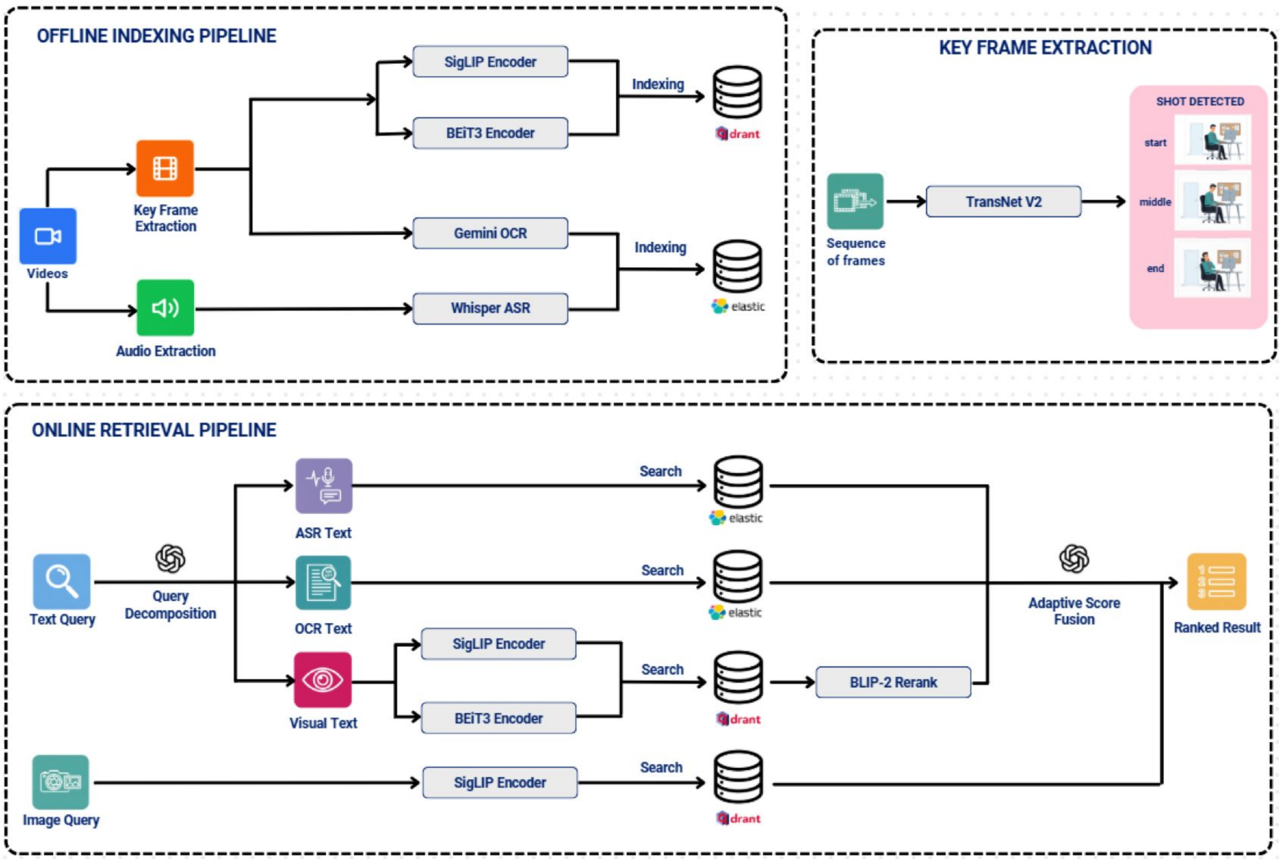
1. Technique

More Sharing, More Gain

❖ Overall System



LINK

1. Technique

❖ Reference

Keyframe Extraction:

1. TransNet V2: An effective deep network architecture for fast shot transition detection
2. AutoShot: A Short Video Dataset and State-of-the-Art Shot Boundary Detection

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Translation Model:

1. M2M100
2. NLLB-200
3. Google Translate API
4. Gemini API

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Vector database:

1. https://superlinked.com/vector-db-comparison
2. https://github.com/RyanCodrai/turbovec.git

1. Technique

❖ Reference

1. A Lightweight Moment Retrieval System with Global Re-Ranking and Robust Adaptive Bidirectional Temporal Search - CVPRw 2025
2. Towards Efficient and Robust Moment Retrieval System: A Unified Framework for Multi-Granularity Models and Temporal Reranking - CVPRw 2025
3. LLandMark: A Multi-Agent Framework for Landmark-Aware Multimodal Interactive Video Retrieval - AAAIw 2026
4. Unified Interactive Multimodal Moment Retrieval via Cascaded Embedding-Reranking and Temporal-Aware Score Fusion - AAAIw 2026
5. Integrated Semantic and Temporal Alignment for Interactive Video Retrieval - SOICT2025
6. MADTempo: An Interactive System for Multi-Event Temporal Video Retrieval with Query Augmentation - SOICT2025

1. Technique

❖ Reference

1. SOICT 2025 proceedings [Link]
2. SOICT 2024 proceedings [Link]
3. SOICT 2023 proceedings [Link]
4. VBS Proceedings [Link]
5. Trecvid Proceedings [Link]

2. Advice

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❖ Some exciting tips

- Textual Query Writing

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- Optimal Strategy (Each member plays a specific role)

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- …..

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3. QA + “Tâm sự”

4. QA

4. QA

THANKS
for
WATCHING